Savitribai Phule Pune University
T.Y.BCA (SEM-V) NEP
Practical Examination OCT / NOV
Subject Code -CA- 303 - PCCP

Subject Name- Lab course on CA-302-PCC (Fundamentals of Artificial Intelligence)

Duration: 3 Hours Maximum Marks: 35

Q1) **A** Write a python program to show proper indentation using if statement. [10 marks]

Q2) **B** Write a python program to implement Water Jug Problem. [20 marks]

OR

Q2) **B** Write a python program to design a Model-Based Agent that maintains an internal state of room cleanliness and decides actions based on both current perception and stored memory. [20 marks]

Q3) Viva [05 marks]

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Duration: 3 Hours Maximum Marks: 35

Q1) **A** Write a python program to implement Tower of Hanoi using recursion. [10 marks]

Q2) **B** Write a python program to implement Mini-Max Algorithm. [20 marks]

OR

Q2) **B** Write a python program that predicts temperature based on the number of icecreams sold using Linear Regression Create Dataframe which contain temperature and number of icecream sold. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a python program to predict if a Person is Eligible to Vote (Age ≥ 18) using simple Logistic Classification. Accept person age from user. [10 marks]

Q2) **B** Write a python program to Implement Alpha-Beta Pruning . [20 marks]

OR

Q2) **B** Write a Python program to Plot a scatter plot of sepal length vs sepal width for the Iris dataset. Label the axes and add a title give appropriate legend. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a python program to simulate the Monkey Banana Problem. Initialize the monkey's position, perform the necessary actions to reach the banana, and use conditional statements to determine whether the goal state has been achieved. [10 marks]

Q2) **B** Apply Linear Regression on a YouTube Video dataset to predict video views using relevant video attributes. Visualize the relationship between the predictor and target variable using a scatter plot and regression line. [20 marks]

OR

Q2) **B** Write a pyhton program to create a simple chatbot using predefined rules to respond to user inputs. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a python program to implement Hill Climbing Algorithm . [10 marks]

Q2) **B** Write a python program to plot a scatter plot of petal length vs petal width for the Iris dataset. Label the axes and add a title. [20 marks]

OR

Q2) **B** Write a python program to implement the Alpha–Beta pruning with depth limit and heuristic. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a python program to simulate the Monkey Banana Problem. Initialize the monkey's position, perform the necessary actions to reach the banana, and use conditional statements to determine whether the goal state has been achieved. [10 marks]

Q2) **B** Write a python program to implement a Logistic Regression classifier for Diabetes Prediction. Train the model using a sample dataset and predict the diabetes status (Diabetic/Non-Diabetic) for multiple persons based on Fasting Blood Sugar, Post-Meal Blood Sugar, Age, and BMI. [20 marks]

OR

Q2) **B** Write a python program for DataFrame of employees with Salary and Experience: If Experience < 3 years, increase salary by 10%, else increase by 5%. and display updated DataFrame. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a program to create a DataFrame which contains a person name and age ,Sort a DataFrame in ascending order of Age [10 marks]

Q2) **B** Write a program to implement the Blocks World Problem. Represent the initial and goal states of a set of blocks, perform valid block movements according to the problem constraints and display each intermediate state until the goal configuration is achieved. [20 marks]

OR

Q2) **B** Write a python program to compare BFS and DFS performance for graph search problems**.** [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a program to separate the features and target variable from a dataset. Split the dataset into training and testing sets display the shapes of the original dataset, training dataset, and testing dataset. Use any suitable dataset. [10 marks]

Q2) **B** Write a python program to solve the N-Queens problem using a backtracking approach. Display one valid arrangement of N queens on an N x N chessboard such that no two queens attack each other [20 marks]

OR

Q2) **B** Write a python program to implement the Tic-Tac-Toe game. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a python program to read the salary_data.csv dataset, train a Linear Regression model using Years of Experience as the feature and Salary as the target and predict the salary for a new employee**.** [10 marks]

Q2) **B** Write a python program to implement the Water Jug Problem. [20 marks]

OR

Q2) **B** Write a python program using Logistic Regression to detect credit card fraud using the following features: Transaction Amount, Card Age (in years), Transaction Time (023),Location Match Score (0–100),Online Transaction (1 = Yes, 0 = No),Train the model and predict fraud (1) or legit (0) based on user input. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a python program to read the car_data.csv dataset and visualize the relationship between Car_Age and Selling_Price using a Scatter Plot. [10 marks]

Q2) **B** Write a python program to implement Tower of Hanoi using recursion. [20 marks]

OR

Q2) **B** Write a python program to simulate a Vacuum Cleaner Agent. The agent should accept the current location and the cleanliness status of the room as input and decide whether to move to another location, clean the room, or remain in the current location. Display the agent's action based on the given conditions. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a Python Program to Split the Iris dataset into training and testing sets (80-20) and print the shapes of X_train and X_test. [10 marks]

Q2) **B** Write a python program to implement a Model-Based Agent for Smart Traffic Signal Control by maintaining previous and current traffic density before deciding the signal status. [20 marks]

OR

Q2) **B** Write a python program to perform Linear Regression using a suitable dataset and predict the output for user-defined input values. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a python program to check whether a given string is a Palindrome. [10 marks]

Q2) **B** Write a python program to implement DataFrame operations on employee records, including sorting, filtering and updating records using the Pandas library. [20 marks]

OR

Q2) **B** Develop a python program to demonstrate the Perception → Decision → Action cycle by simulating an Intelligent Agent in a simple environment. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a program to create a DataFrame which contains a person name and age ,Sort a DataFrame in ascending order of Age [10 marks]

Q2) **B** Write a python program to load the Diabetic in built dataset, convert it into a Data frames and display the first 7 rows and the shape of the dataset. [20 marks]

OR

Q2) **B** Write a python program to using Logistic Regressions to detect credit card fraud using the following features: Transaction Amount ,Card Age(calculated into years), Transaction Time(0-23),Location Matching scores(0-100),Online Transcation (0=NO,1=Yes)Train the model and predict fraud(1) or legit (0) based on user input. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a python program to check whether a given number is a Prime Number using a user-defined function. [10 marks]

Q2) **B** Write a python program to load employee.csv containing Employee ID, Salary, and Department dataset and perform basic data exploration. [20 marks]

a.Display the first four rows of the dataset.
b. Print the shape of the dataset.
c. Show column names.
d. Identify missing values.
e. Display dataset information.

OR

Q2) **B** Write a python program that predicts the price of a house using a dataset that includes Area (sq.ft) and Number of Bedrooms as input features from the user. Use Multiple Linear Regression to build the prediction model. [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a Python program to print "Welcome to Artificial Intelligence". [10 marks]

Q2) **B** Write a python program using Decision Tree Classification to predict the type of vegetables (Leafy vegetables or fruit vegetabels) based on the following features: it is leafy vegetabels or not, how to grow vegetables : grow from flower = 1,otherwise 0 [20 marks]

OR

Q2) **B** Write a python program to develop fitness predictors using Random Forest Classification to predict whether a person is FIT or UNFIT based on: Age, Exercise Hours per Week Accept the input from user and predict for multiple people [20 marks]

Q3) Viva [05 marks]

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Q1) **A** Write a python program to count the frequency of each word in a given string using a dictionary. [10 marks]

Q2) **B** Write a python program to create a simple chatbot using predefined rules to respond to user inputs. [20 marks]

OR

Q2) **B** Write a Python program using Decision Tree Classification to predict the type of fruit(Apple or Orange) based on the following features: Weight of the fruit (in grams) ,42 Texture of the fruit : Smooth = 0 , Rough = 1. [20 marks]

Q3) Viva [05 marks]